SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEIRING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Experiments

Course Code:	ITA05	**Course Title:**	Computer Vision
CO Assessed:	CO3 – Precision (accurate feature extraction & analysis) (BL3)	**Assessment:**	Assessment Tool 1 – Experiments
Weightage:	40% of CIA	**Total Marks:**	100
CO3 Statement:	*Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BTL 4)*		
Student Name:	G. Tejasree	**Reg. No.:**	192511204
Section / Batch:	CSE A	**Date:**	August 13 2026

Code of Conduct

I, G. Tejasree (Name / Reg No) *certify that this submission is my original work and that I adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: G. Tejasree

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts	/ 25
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation	/ 30
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools	/ 20
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results	/ 15
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation	/ 10
					Total Marks	**/ 100**

edge_detection_comparison.ipynb

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```
import cv2
import matplotlib.pyplot as plt

# Read image
img = cv2.imread('building.jpg')
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Sobel Edge Detection
sobel_x = cv2.Sobel(gray, cv2.CV_64F, 1, 0, ksize=3)
sobel_y = cv2.Sobel(gray, cv2.CV_64F, 0, 1, ksize=3)
sobel = cv2.magnitude(sobel_x, sobel_y)

# Canny Edge Detection
canny = cv2.Canny(gray, 100, 200)

# Display results
plt.figure(figsize=(12,4))

plt.subplot(1,3,1)
plt.imshow(gray, cmap='gray')
plt.title('Original Image')
plt.axis('off')

plt.subplot(1,3,2)
plt.imshow(sobel, cmap='gray')
plt.title('Sobel Edge Detection')
plt.axis('off')

plt.subplot(1,3,3)
plt.imshow(canny, cmap='gray')
plt.title('Canny Edge Detection')
plt.axis('off')

plt.show()
```

Original Image　　Sobel Edge Detection　　Canny Edge Detection

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